

Design doc — Week 2

Mandatory for mentor review. Open a GitHub Issue linking to this file before Week 3.

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Repo: https://github.com/Sereen-Karakra/nextflows_project
Branch: [week-2-design](#)
GitHub Issue: https://github.com/Sereen-Karakra/nextflows_project/issues/1

1. Pitch

Notes & FAQ Search is an MCP server designed to help students quickly search through notes and frequently asked questions. Instead of manually browsing documents, users can ask questions in natural language and receive relevant answers. The MCP exposes simple tools for searching notes, listing available notes, and adding new notes, making information easier to access during conversations.

2. Demo Day user story

1. The mentor starts the MCP Inspector and connects to the Notes & FAQ Search MCP server.
2. The user asks: "Can you find notes about Git branches?"
3. The model calls the [search_notes](#) tool to search for matching notes. If needed, it can call [list_notes](#) to show the available notes.
4. The audience sees the matching note returned with a useful answer, demonstrating how the MCP server helps retrieve information quickly.

3. Tool inventory (4–7 tools)

Mark exactly **three** tools as **P0** (must work for Demo Day). Others can be P1 stubs.

Priority	Tool name (verb_noun)	Description (for the model)	Inputs	Outputs
P0	search_notes	Search notes using keywords and return matching results.	keyword	Matching notes
P0	list_notes	Return a list of all available notes.	None	List of note titles
P0	add_note	Add a new note to the collection.	title, content	Success message
P1	update_note	Update an existing note.	title, new content	Updated note
P1	delete_note	Delete a note from the collection.	title	Success or error message

4. Out of scope

The following features are not included in this project:

- User authentication and account management.
- Mobile or web user interface.
- Integration with paid APIs or cloud database services.

5. Success criteria

You succeed on Demo Day if:

- ☐ The model successfully searches notes and returns relevant results.
- ☐ The model lists the available notes correctly.
- ☐ The model adds a new note successfully using the MCP tool.

6. Top risks

Risk	Likelihood	Mitigation
Limited time to complete all planned features	Medium	Focus on implementing all P0 tools before adding extra features.
MCP tool integration issues	Medium	Test each tool individually using MCP Inspector before the demo.
Incorrect search results	Low	Test the search tool with multiple sample notes and improve keyword matching.

7. Evidence for Week 2

- ☒ docs/project-choice.md filled
- ☐ ≥3 Zod schemas under src/schemas/
- ☐ Tools registered (stubs OK)
- ☐ examples/<tool>.json for each registered tool
- ☐ Inspector screenshots attached to the GitHub Issue

Mentor decision

- Status: pending
- Comments: